Data Analytics Capstone: Analysis of Facility and Unit Attributes under the EPA Clean Air Markets Program Using Statistical and Machine Learning Techniques

Interim Capstone Report

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QM640 V1: Data Analytics Capstone
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Summer 2026 Term
October 15, 2026

Executive Summary & Project Repository

This Interim Capstone Report is submitted as a standalone academic document in partial fulfillment of the requirements for the Master of Science in Data Analytics and Artificial Intelligence at Walsh College. In accordance with the academic curriculum guidelines of QM640 V1, this report provides an autonomous, self-contained evaluation of the research design, empirical data cleaning, exploratory data analysis (EDA), statistical hypothesis testing, and preliminary machine learning modeling conducted on electric power generating facilities regulated under the United States Environmental Protection Agency (EPA) Clean Air Markets Program (CAMP).

GitHub Data Availability Statement

To support scientific reproducibility, public auditability, and academic evaluation, all raw data artifacts, engineered datasets, Python modeling scripts, statistical pipelines, and visualization routines have been systematically curated and deposited in the official project repository:

Project Repository: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone

Raw Data URL: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/blob/main/data/raw/facility-2025.csv

Processed Data URL: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/blob/main/data/processed/facility_clean_processed.csv

Code/Notebooks: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/tree/main/scripts

Figure Directory: outputs/figures/ (10 APA 7 publication-grade figures at 300 DPI)

> **Academic Standing Declaration:** *Standalone Academic Notice: While this Interim Report logically succeeds the initial Capstone Synopsis submitted in the Winter term, no prior dependency or external knowledge of the synopsis is required. A reader, evaluator, or regulatory auditor can comprehensively evaluate the theoretical foundation, analytical methodology, empirical results, and strategic implications from this standalone document alone.*

1. Introduction

Background and Context

Atmospheric pollution originating from stationary combustion sources constitutes one of the most critical public health and environmental challenges of the modern industrial era. Electric power generation facilities—primarily those combusting fossil fuels including bituminous coal, sub-bituminous coal, pipeline natural gas, and residual fuel oils—are major point sources of criteria air pollutants. These combustion processes emit substantial quantities of sulfur dioxide (SO_2), nitrogen oxides (NO_x), hazardous air pollutants such as mercury (Hg), and primary particulate matter ($PM_{2.5}$ and PM_{10}). Epidemiological and economic studies have repeatedly demonstrated that ambient exposure to fine particulates and ground-level ozone causes severe cardiorespiratory illnesses, premature mortality, and regional ecological acidification (Harrington et al., 2004; Burtraw et al., 2012).

To curtail stationary source emissions, the United States Environmental Protection Agency (EPA) established market-based regulatory frameworks under the Clean Air Act Amendments of 1990. These

initiatives encompass the Acid Rain Program (ARP), the Clean Air Interstate Rule (CAIR), the Cross-State Air Pollution Rule (CSAPR, including CSNOX and CSSO2G2), and the Mercury and Air Toxics Standards (MATS). Central to these market-oriented programs is the Clean Air Markets Program (CAMP), which couples statutory emission allowance trading with rigorous, continuous emission monitoring systems (CEMS) mandated under 40 CFR Part 75 (U.S. EPA, 2024). Under this regime, the EPA collects and publishes unit-level data capturing thermal capacities, primary and secondary fuel choices, boiler architectural configurations, and pollution abatement controls.

The operational setting of this research is the contemporary U.S. bulk power system during the calendar year 2025. The electric power sector is currently undergoing a structural transformation characterized by the rapid retirement of legacy coal-fired generating units, increased utilization of high-efficiency natural gas combined cycle (NGCC) plants, the deployment of fast-ramping combustion turbines for renewable intermittency balancing, and the impending enforcement of stricter environmental regulations (Ellis et al., 2019; Schmalensee & Stavins, 2017). Within this setting, primary stakeholders include federal environmental regulators (EPA program administrators), regional transmission organizations (RTOs and ISOs), state environmental protection agencies, electric utility asset managers, and ESG investment analysts who require quantitative, data-driven models to benchmark emission compliance risk and project fleet transition trajectories.

Problem Statement

Despite decades of standardized environmental regulations and mandatory reporting under the Clean Air Markets Program, operational thermal capacity and potential emission rates exhibit pronounced heterogeneity across the national generating fleet. Existing compliance frameworks frequently treat generating units as homogeneous technological entities within broad fuel categories, overlooking complex multidimensional interactions among boiler vintage, unit age, installed generation capacity, pollution control architectures, and regional regulatory enforcement. The absence of scalable, data-driven predictive and segmentation models limits the ability of environmental regulators to anticipate high-emission non-compliance risks proactively and impedes power plant operators from establishing empirical benchmarks for unit retrofitting or retirement. This research addresses this gap by synthesizing statistical inferential modeling with supervised and unsupervised machine learning to explain, predict, and segment facility-level operational intensity and compliance risk.

Purpose of the Study

The primary purpose of this quantitative empirical study is fourfold: (a) to evaluate whether facility attributes—specifically primary fuel classification and boiler design—exert statistically significant effects on maximum hourly heat input rate and emission capacity; (b) to construct and validate supervised classification algorithms capable of predicting high-emission generating units based on operational features; (c) to assess the linear and non-linear bivariate and multivariate relationships between generating unit vintage (age) and thermal operational intensity; and (d) to formulate an unsupervised clustering framework that segments the U.S. power generation fleet into distinct, interpretable operational and emission-risk archetypes.

Interim Project Status (Progress Snapshot)

A structured summary of completed, in-progress, and pending milestones is outlined below:

Completed: Data acquisition from EPA CAMPD bulk repository (N = 4,050 records, 30 variables); data cleaning pipeline; imputation strategy; duplicate validation; feature engineering of Unit Age, Capacity (MWe), Heat Input Rate, and Pollution Control counts; exploratory data analysis; statistical testing for RQ1 (ANOVA, Kruskal-Wallis, Tukey HSD); exploratory scatter and correlation matrices.

In Progress: Supervised machine learning model training and hyperparameter optimization (Logistic Regression, Random Forest, Gradient Boosting) for RQ2; OLS simple and multivariate regression modeling for RQ3; K-Means unsupervised clustering, Elbow WCSS optimization, Silhouette coefficient evaluation, and 2D Principal Component Analysis (PCA) for RQ4.

Pending: Longitudinal panel data expansion incorporating 2020-2024 historical CEMS hourly emissions; integration of state-level ambient air quality monitoring stations; deployment of SHAP (Shapley Additive exPlanations) values for global/local machine learning interpretability; formulation of prescriptive policy and retrofitting guidelines for the final capstone deliverable.

Scope and Objectives

The spatial scope of this investigation encompasses all stationary electric utility and industrial power generating units located in the 50 U.S. states, the District of Columbia, and associated territories reporting under the EPA Clean Air Markets Program for the 2025 reporting period. The temporal scope focuses on cross-sectional operational attributes registered as of 2025. The study explicitly focuses on operational combustion parameters, installed capacity, pollution abatement equipment, and regulatory classifications. Detailed dispersion plume modeling, downstream meteorological transport simulations, human health epidemiological impact modeling, and corporate balance sheet financial accounting fall outside the immediate scope of this empirical investigation.

Research Questions and Hypotheses

To provide rigorous quantitative resolution to the problem statement, four formal research questions (RQs) and associated statistical hypotheses are investigated:

Research Question 1 (RQ1): Emission & Thermal Load Variability across Fuels

Do facility attributes, specifically primary fuel category and unit technology type, significantly influence maximum hourly heat input rates and operational combustion capacity across the U.S. power fleet?

H0: Null Hypothesis (H0): There is no statistically significant difference in mean or median maximum hourly heat input rates across primary fuel categories ($\mu_Coal = \mu_Gas = \mu_Oil = \mu_Biomass$).

H1: Alternative Hypothesis (H1): Maximum hourly heat input rates differ significantly across primary fuel categories, with fossil solid fuels exhibiting systematically higher combustion capacities.

Analytical Formulation: Statistical Model: $Heat_Input_Rate_{ij} = \mu + \alpha_i (Fuel_Category) + \varepsilon_{ij}$, evaluated via One-Way Analysis of Variance (ANOVA) and non-parametric Kruskal-Wallis H-testing.

Research Question 2 (RQ2): Predictive Classification of High-Emission Units

Can facility and unit operational attributes (installed capacity, boiler vintage, pollution control count, and regional location) accurately classify and predict high-emission generating units (defined as units operating in the upper quartile, ≥ 75th percentile of national maximum hourly heat input capacity)?

H0: Null Hypothesis (H0): Operational and technological attributes possess no predictive power (AUC ≤ 0.50, F1-score equal to random guessing) in classifying high-emission generating units.

H1: Alternative Hypothesis (H1): Machine learning classifiers trained on unit attributes can predict high-emission facilities with superior discriminative performance (ROC-AUC ≥ 0.90, F1-score ≥ 0.80).

Analytical Formulation: Model Specifications: Regularized Logistic Regression, Random Forest Ensembles (MDI Gini splitting), and Gradient Boosting Classifiers evaluated via 5-fold stratified cross-validation.

Research Question 3 (RQ3): Unit Age and Operational Intensity Dynamics

What is the empirical nature, magnitude, and statistical significance of the relationship between generating unit age (vintage) and maximum thermal heat input capacity?

H0: Null Hypothesis (H0): There is no statistically significant linear or monotonic relationship between generating unit age and maximum hourly heat input rate ($\beta_{Age} = 0$; Pearson $r = 0$; Spearman $\rho = 0$).

H1: Alternative Hypothesis (H1): A statistically significant positive relationship exists between unit age and thermal heat input rate ($\beta_{Age} > 0$), reflecting the larger combustion volume of legacy baseload steam facilities.

Model Equations: Analytical Formulation: Bivariate Ordinary Least Squares (OLS): $Heat_Input = \beta0 + \beta1(Age) + \varepsilon$; Multivariate OLS: $Heat_Input = \beta0 + \beta1(Age) + \beta2(Capacity_MW) + \beta3(Control_Count) + \varepsilon$.

Research Question 4 (RQ4): Unsupervised Operational Segmentation & Archetypes

Can the heterogeneous national generating fleet be partitioned into distinct, objectively validated operational clusters based on continuous attributes of capacity, thermal input, unit age, and pollution control adoption?

H0: Null Hypothesis (H0): Power generation units form a continuous, undifferentiated operational distribution with no identifiable, cohesive clustering structure (Silhouette score ≤ 0.20).

H1: Alternative Hypothesis (H1): Generating facilities naturally cluster into distinct, interpretable operational archetypes exhibiting substantial between-cluster separation and within-cluster cohesion.

Clustering Framework: Analytical Formulation: K-Means clustering across standardized multidimensional space; cluster count selection via Within-Cluster Sum of Squares (Inertia Elbow) and Silhouette coefficient optimization; 2D projection via Principal Component Analysis (PCA).

2. Literature Survey

Literature Review Approach

A systematic literature review was conducted to synthesize empirical methodologies, economic theories, and environmental engineering principles relevant to facility-level emissions, air pollution regulatory compliance, and machine learning applications in energy systems. Electronic databases queried included IEEE Xplore, ScienceDirect, Elsevier Energy Policy, SpringerLink, JSTOR, and Google Scholar. The query syntax utilized Boolean logic: ('Clean Air Markets Program' OR 'Acid Rain Program' OR 'CAMPD') AND ('facility attributes' OR 'boiler vintage' OR 'heat input') AND ('machine learning' OR 'clustering' OR 'classification' OR 'ANOVA').

Inclusion criteria required peer-reviewed journal articles, foundational statistical texts, or authoritative government reports published between 1952 and 2024 that addressed stationary combustion sources, environmental policy instruments, non-parametric hypothesis testing, regression diagnostics, or industrial clustering. Theoretical publications lacking empirical grounding or studies restricted entirely to mobile vehicular emissions were excluded. A total of 14 seminal sources were identified and synthesized.

Summary of Key Literature

The 14 foundational sources are itemized in Table 1, summarizing author credentials, publication year, operational domain, underlying datasets, analytical methods, empirical findings, and direct linkage to the research questions investigated in this capstone.

Table 1

Literature Relevance Matrix

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	Linkage to RQs
U.S. EPA (2024)	Environmental Regulation	EPA CAMPD Bulk Files (2024-2025)	Continuous Emission Monitoring (CEMS), 40 CFR 75	Transparency in reporting drives compliance; fuel and boiler design govern baseline emissions.	Provides raw data schema, unit-level dictionary, and regulatory baseline for RQ1-RQ4.
Ellis et al. (2019)	Energy & Environmental Policy	U.S. Power Plant Panel (2000-2016)	Fixed Effects Econometrics, Panel Regression	Market-based programs induced massive fuel switching from coal to natural gas, reducing SO2 by >80%.	Supports fuel category grouping (RQ1) and feature selection for predictive classification (RQ2).
Schmalensee & Stavins (2017)	Resource Economics	3 Decades of Cap-and-Trade Systems	Policy Retrospective, Cost-Benefit Analysis	Allowance trading achieved emission reductions at a fraction of command-and-control costs.	Theoretical justification for regulatory program codes (ARP, CSNOX, MATS) as predictive features.
Burtraw et al. (2012)	Electricity Market Modeling	U.S. Electric Reliability Councils	Linear Programming, Simulation Modeling	Fleet reliability under carbon/emission constraints depends heavily on unit age and ramping speed.	Directly links unit vintage and thermal heat input rate to operational risk (RQ3).
Harrington et al. (2004)	Environmental Policy Instruments	Comparative U.S. and European Fleets	Comparative Institutional Analysis	Economic incentives outperform prescriptive standards only when continuous monitoring is robust.	Justifies using continuous heat input and control counts as objective compliance indicators.
Hollander et al. (2014)	Nonparametric Statistics	Textbook & Reference Compendium	Kruskal-Wallis, Wilcoxon-Mann-Whitney Tests	Rank-based non-parametric tests maintain statistical power when normality assumptions fail in	Direct methodological framework for non-parametric Kruskal-Wallis testing in RQ1.

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	Linkage to RQs
				skewed data.	
Gibbons & Chakraborti (2011)	Statistical Inference	Foundational Mathematical Text	Distribution-Free Inferences, Effect Sizes	Formulated asymptotic relative efficiency and rank transformation mechanics for heavy-tailed data.	Provides mathematical rigor for post-hoc pairwise multiple comparison adjustments.
Montgomery et al. (2021)	Econometrics & Regression	Industrial Engineering Applications	OLS, Multicollinearity VIF, Residual Diagnostics	Outlines Gauss-Markov assumptions, heteroscedasticity remedies, and multivariate regression.	Methodological blueprint for bivariate and multivariate regression modeling in RQ3.
Field (2018)	Applied Data Analytics	Statistical Modeling Handbook	ANOVA, Effect Size (Eta-Squared), Post-Hoc Tests	Detailing ANOVA variance decomposition, SS_between, SS_within, and Tukey HSD test formulations.	Supplies calculation formulas for ANOVA F-statistic and Eta-squared effect size in RQ1.
Cochran (1977)	Sampling Theory	Mathematical Statistics Reference	Stratified Sampling, Power Analysis	Established sample size lower bounds for group comparison and variance estimation across strata.	Validates sample size adequacy (N = 4,050 exceeds minimum required sample of 180).
Kruskal & Wallis (1952)	Mathematical Statistics	Original Methodological Derivation	H-statistic Rank Sum Derivation	Proved asymptotic chi-square distribution of rank sums for k independent groups.	Primary mathematical origin of the non-parametric H-test utilized for RQ1.
Breiman (2001)	Machine Learning	Algorithm Derivation	Random Forests, Bagging, Out-of-Bag Error	Ensemble bagging over decorrelated decision trees minimizes variance and prevents overfitting.	Methodological justification for Random Forest classifier implementation in RQ2.
Friedman (2001)	Supervised Learning	Algorithmic Development	Gradient Boosting Machine, Stage-wise Modeling	Sequential gradient descent in function space optimizes complex non-linear classification loss.	Theoretical basis for Gradient Boosting benchmark classifier in RQ2.
MacQueen (1967)	Unsupervised Learning	Clustering Theory	K-Means Centroid Optimization	Iterative partitioning minimizing within-cluster sum of squares converges to local Pareto optima.	Algorithmic foundation for unsupervised fleet segmentation and elbow method in RQ4.

Note. Synthesized literature base establishing theoretical, mathematical, and algorithmic foundations for RQ1-RQ4.

Thematic Synthesis of Literature

Theme 1: Policy Mechanisms and Fuel Transition Dynamics. The literature on market-based emissions trading (Schmalensee & Stavins, 2017; Ellis et al., 2019) overwhelmingly documents that statutory cap-and-trade programs fundamentally altered the merit-order dispatch of electric power plants. By establishing explicit allowance prices for SO_2 and NO_x, the Clean Air Act amendments rendered unscrubbed coal plants economically uncompetitive relative to combined cycle natural gas units. Consequently, primary fuel selection became the primary determinant of both thermal operational capacity and absolute emission mass. This finding directly motivates RQ1, which quantitatively tests the degree to which fuel categories maintain distinct heat input and operational capacity profiles in 2025.

Theme 2: Generating Unit Vintage, Size, and Abatement Asymmetry. A recurring observation across environmental economics and power systems engineering (Burtraw et al., 2012; Harrington et al., 2004) is the 'grandfathering' of legacy generating assets. Because statutory new source performance standards (NSPS) imposed stringent control requirements exclusively on newly constructed facilities, electric utilities were economically incentivized to prolong the operating lifespans of aging, large-scale steam boilers built prior to 1980. This created a dual-regime fleet: older, very large baseload units equipped with capital-intensive post-combustion scrubbers (flue gas desulfurization [FGD] and selective catalytic reduction [SCR]) coexisting with modern, highly efficient, smaller combustion turbines. This structural asymmetry motivates the regression modeling in RQ3 and the multidimensional clustering framework in RQ4.

Theme 3: Statistical Rigor in Environmental Data Analytics. Foundational statistical literature (Hollander et al., 2014; Field, 2018; Montgomery et al., 2021) strongly cautions against relying exclusively on parametric linear models when analyzing industrial operational data. Facility-level heat input rates and installed capacities invariably exhibit extreme right-skewness, heavy tails, and heteroscedastic variance across fuel groupings. Accordingly, this capstone couples parametric One-Way ANOVA with non-parametric Kruskal-Wallis rank tests and complements bivariate Pearson correlations with Spearman rank correlations, ensuring inferential validity under non-Gaussian distributional conditions.

Theme 4: Machine Learning for Fleet Risk Classification. In modern energy data science, ensemble machine learning methods (Breiman, 2001; Friedman, 2001) offer substantial advantages over traditional parametric models by capturing non-linear interactions between regulatory program participation, physical unit size, and geographical region without restrictive distributional assumptions. However, literature highlights the necessity of rigorous validation—specifically stratified cross-validation and ROC-AUC assessment—to avoid misleading accuracy metrics in the presence of class imbalance. These principles directly guide the supervised learning architecture developed for RQ2.

3. Data Description

Data Source(s) and Access

The empirical dataset analyzed in this capstone is sourced directly from the U.S. Environmental Protection Agency (EPA) Clean Air Markets Program Data (CAMPD) bulk data archive. CAMPD constitutes the official federal repository for compliance, facility attribute, and emission monitoring data collected under Title IV and Title I of the Clean Air Act. The data were accessed via the official EPA bulk data portal:

Primary Data Source: URL: https://campd.epa.gov/data/bulk-data-files (Dataset: Facility/Unit Attributes – 2025)

Citation: APA 7 Citation: U.S. Environmental Protection Agency. (2024). Clean Air Markets Program data: Bulk data files [Data set]. U.S. EPA. https://campd.epa.gov/data/bulk-data-files

Dataset Overview

The raw dataset consists of a single cross-sectional tabular file (facility-2025.csv) containing unit-level technical and administrative attributes for power generating facilities across the United States. Key summary parameters include:

Number of Records (Rows, N): 4,050 generating units across 1,328 unique facilities

Number of Variables (Columns, p): 45 attributes (30 original CAMPD variables + 15 engineered features)

Temporal Coverage: Calendar Year 2025 (covering cross-sectional active status and operating dates)

Unit of Analysis: Electric power generating unit (boiler, combustion turbine, combined cycle train)

Geographic Scope: 52 jurisdictions (50 U.S. states, District of Columbia, and Puerto Rico/Territories)

Target Variables: High_Emitter (Binary indicator: 1 = Max Hourly HI Rate ≥ 2350.0 mmBtu/hr, 75th percentile; 0 otherwise) and Heat_Input_Rate (Continuous combustion capacity, mmBtu/hr).

Data Dictionary

Table 2 provides the mandatory data dictionary detailing the core raw and engineered variables, definitions, storage datatypes, allowed ranges, missing value handling strategies, and relevant analytical notes.

Table 2

Data Dictionary for Clean Air Markets Program Facility Attributes

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Handling	Analytical Notes
State	U.S. state postal code of facility	String (Categorical)	52 states/territories	None (0 missing)	Identifies state-level environmental jurisdiction.
Facility ID	Unique federal identifier assigned by EPA / ORISPL	Integer	3 to 67540	None (0 missing)	Primary key for facility-level aggregation.
Facility Name	Official registered plant or facility name	String	1,325 unique names	None (0 missing)	Commercial plant designation.
Unit ID	Internal unit or boiler identifier	String	1,412 unique unit tags	None (0 missing)	Disambiguates units within multi-unit facilities.
Unit Type	Boiler combustion or turbine technology	String (Categorical)	33 categories	None (0 missing)	Recoded into 5 major technology classes.
Primary Fuel Type	Main combustible fuel utilized for energy	String (Categorical)	12 fuel codes	44 missing (Imputed to Other/Unknown)	Recoded into 5 primary fuel categories for RQ1.

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Handling	Analytical Notes
Associated Generators & Capacity	Generator name and nameplate electrical capacity (MWe)	String (Formatted text)	Text e.g. '1 (153.1)'	150 missing (Parsed and imputed via median)	Parsed via regex to extract total installed Capacity_MW.
Max Hourly HI Rate (mmBtu/hr)	Maximum design hourly thermal heat input rate	Float (Continuous)	94.0 to 16,250.0 mmBtu/hr	44 missing (Imputed via Capacity ratio)	Primary metric of operational combustion intensity.
Commercial Operation Date	Date the unit commenced commercial generation	Date / String	1/1/1930 to 11/23/2025	36 missing (Imputed via unit-type median)	Converted to Unit_Age = 2025 - Year.
Operating Status	Current operational status in 2025	String (Categorical)	Operating, Retired, Future	None (0 missing)	Recoded into binary Is_Operating flag.
SO2 Controls	Installed sulfur dioxide abatement systems	String (Multi-valued)	Wet Limestone, Dry FGD, etc.	3,678 missing (Interpreted as Uncontrolled)	Recoded to binary flag Has_SO2_Control.
NOx Controls	Installed nitrogen oxide abatement systems	String (Multi-valued)	Low NOx Burners, SCR, SNCR	596 missing (Interpreted as Uncontrolled)	Recoded to binary flag Has_NOx_Control.
PM Controls	Installed particulate matter abatement equipment	String (Multi-valued)	Electrostatic Precipitator, Baghouse	3,540 missing (Interpreted as Uncontrolled)	Recoded to binary flag Has_PM_Control.
Hg Controls	Installed mercury abatement technologies	String (Multi-valued)	Activated Carbon Injection, etc.	3,877 missing (Interpreted as Uncontrolled)	Recoded to binary flag Has_Hg_Control.
Capacity_MW	Engineered total installed generation capacity	Float (Continuous)	5.0 to 1,300.0 MWe	Engineered from regex parsing	Key predictor in RQ2 and covariate in RQ3.
Unit_Age	Engineered unit operating vintage in years	Float (Continuous)	0.0 to 95.0 years	Engineered from commercial date	Primary independent variable for RQ3.
Control_Count	Total number of installed pollution abatement systems	Integer (Discrete)	0 to 4 installed systems	Engineered sum of binary flags	Captures overall environmental control investment.
High_Emitter	Binary indicator of top quartile heat input rate	Integer (Binary: 0, 1)	0 (Low/Med), 1 (High Emitter)	Engineered via 75th percentile cutoff	Supervised classification target for RQ2.

Note. Comprehensive data dictionary detailing raw CAMPD attributes and engineered analytical features.

Sample Size Justification & Analytical Record Allocation across Models

A foundational methodological strength of this capstone research is that modeling and statistical inferences were conducted on the complete national fleet of 4,050 generating units across 1,328 unique facilities, representing 100% of regulated units in facility-2025.csv with zero records discarded. In accordance with the sampling methodology established in the capstone synopsis, statistical power calculations (Cohen, 1988; Cochran, 1977) confirm that the 4,050 records exceed the minimum sample size requirements for each analytical model by an order of magnitude.

For RQ1, group comparisons across fuel categories require a minimum of 30 observations per group (rule of thumb) and an aggregate sample of 159–180 units for medium effect size ($f = 0.25$, power = 80%, $\alpha = 0.05$); the available 4,050 units exceed this threshold by over 22-fold. For RQ2, predictive classification guidelines dictate 10–20 observations per candidate predictor per class (minimum 120–240 units for 6–12 predictors); our stratified partitioning allocates 3,240 records (80%) to the training set with 5-fold cross-validation and preserves 810 records (20%) as a holdout test set (containing 204 actual high emitters and 606 low emitters), ensuring exceptional model stability and preventing data leakage. For RQ3, linear and multivariate regressions require 300–800 records for small-to-medium effect detection; all 4,050 units are utilized (degrees of freedom = 4,046). For RQ4, unsupervised clustering heuristics require at least 10× the feature count (≥ 300 units); K-Means is executed across the entire 4,050-unit fleet. Table 3 details the exact record allocation across all models.

Table 3

Analytical Model Record Allocation and Sample Size Justification

Research Question & Scope	Analytical Model	Total Records	Training Set	Validation / Holdout	Minimum Needed (Power)	Statistical Power Evaluation
RQ1: Emission & Heat Input Variability across Fuel Types	One-Way ANOVA & Kruskal-Wallis Test	4,050 records (100%)	4,050 records (Full population)	N/A (Omnibus inference)	159–180 units ($f = 0.25$, power = 80%)	Exceeds minimum by >22×; highly robust to non-normality.
RQ2: Predictive Classification of High-Emission Units	Random Forest, Gradient Boosting, Logistic Regression	4,050 records (100%)	3,240 records (80% stratified)	810 records (20% holdout: 204 High / 606 Low)	120–240 units (10–20 per predictor/class)	Exceeds minimum by >13×; 5-fold CV ensures generalizability without leakage.
RQ3: Unit Age and Operational Thermal Intensity	Bivariate & Multivariate OLS Regressions	4,050 records (100%)	4,050 records (Full population)	N/A (df = 4,046; residual diagnostics)	300–800 units (Small-medium effect, $r \approx 0.1$-0.3)	Exceeds minimum by >5×; highly stable parameter standard errors.
RQ4: Unsupervised Fleet Segmentation & Archetype Profiling	K-Means Clustering & 2D Principal Component Analysis	4,050 records (100%)	4,050 records (Full population)	N/A (Fleet-wide centroid optimization)	≥ 300 units (≥ 10× feature count rule)	Exceeds minimum by >13×; stable centroid formation and silhouette.

Note. Comprehensive allocation of records across models confirming 100% data preservation (zero records discarded) and statistical power adequacy.

GitHub Data Availability Statement

In accordance with university capstone requirements, the complete raw and processed datasets and Python notebooks/scripts are permanently maintained and available in the GitHub repository under:

Raw data: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/blob/main/data/raw/facility-2025.csv

Processed/clean data: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/blob/main/data/processed/facility_clean_processed.csv

Code/notebooks: https://github.com/djmsritai-tech/QM640-Data-Analytics-Capstone/tree/main/scripts

Screenshots / Evidence (as required)

To satisfy the academic audit requirement, the file and directory architecture of the capstone project repository is structured hierarchically as depicted below:

Repository Directory Architecture:

```
QM640-Data-Analytics-Capstone/
│
├── data/
│   ├── raw/
│   │   └── facility-2025.csv            [Raw EPA CAMPD dataset: 4,050 rows, 30 columns]
│   └── processed/
│       └── facility_clean_processed.csv  [Engineered dataset: 4,050 rows, 45 columns]
│
├── scripts/
│   ├── run_capstone_analytics.py        [End-to-end Python statistical & ML pipeline]
│   ├── build_full_interim_report.py     [APA 7 Interim Report generation script]
│   └── build_full_final_report.py       [APA 7 Final Report generation script]
│
├── outputs/
│   ├── analytics_results.json           [Complete JSON export of test statistics & metrics]
│   └── figures/
│       ├── figure0_system_architecture.png
│       ├── figure1_fuel_emission_boxplot.png
│       ├── figure2_unit_type_distribution.png
│       ├── figure3_controls_by_fuel.png
│       ├── figure4_correlation_heatmap.png
│       ├── figure5_age_vs_heat_input_scatter.png
│       ├── figure6_roc_curves.png
│       ├── figure7_confusion_matrices.png
│       ├── figure8_feature_importance.png
│       ├── figure9_kmeans_elbow_silhouette.png
│       └── figure10_pca_clusters.png
│
└── reports/
    ├── QM 640_Interim_Report_DhananjayVerma-0291493.docx & .pdf
    └── QM 640_Final_Report_DhananjayVerma-0291493.docx & .pdf
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4. Analysis & Exploratory Results

Data Cleaning and Preprocessing Log

Rigorous data cleaning and preprocessing protocols were instituted to resolve data quality anomalies, structural formatting irregularities, and missing values inherent in regulatory filings. Table 4 enumerates

the complete data cleaning log, specifying the anomaly detected, variables impacted, detection methodology, corrective treatment applied, and theoretical rationale.

Table 4

Data Cleaning and Preprocessing Log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Duplicate Records	Facility ID + Unit ID	Pandas duplicated() key check	Verified 0 duplicate primary keys across 4,050 rows	Ensures data integrity and prevents artificial sample inflation.
Missing Pollution Controls	SO2, NOx, PM, Hg Controls	isna().sum() check (596 to 3,877 missing)	Imputed as 'Uncontrolled' (0); encoded binary presence flags	In EPA CAMPD, missing control fields denote absence of dedicated abatement hardware.
String-Formatted Capacity	Associated Generators & Nameplate Capacity	Regex search for parenthetical MWe numbers	Extracted numerical MWe values and summed multi-generator units	Converts unstandardized text strings into continuous numerical capacity (Capacity_MW).
Missing Nameplate Capacity	Capacity_MW	150 missing values post-parsing (3.70%)	Imputed using median capacity grouped by Unit_Type_Category	Preserves sample size while avoiding bias from cross-technology capacity disparities.
Commercial Date Formatting	Commercial Operation Date	pd.to_datetime() conversion check	Converted valid dates; 36 missing imputed via unit-type median	Enables exact calculation of continuous unit vintage: Unit_Age = 2025 - Year.
Missing Heat Input Rate	Max Hourly HI Rate (mmBtu/hr)	44 missing values (1.09%)	Imputed using median Heat-Input-to-Capacity ratio for unit type	Maintains physical thermal consistency (nominal heat rate ~ 9.5 mmBtu/MWe).
High-Cardinality Fuel Types	Primary Fuel Type	value_counts() check (12 distinct codes)	Consolidated into 5 robust classes: Natural Gas, Coal, Oil/Diesel, Biomass, Other	Satisfies statistical power requirements (Rule of thumb ≥ 30 units per group for ANOVA).
High-Cardinality Boiler Types	Unit Type	value_counts() check (33 distinct codes)	Aggregated into 5 primary technologies: Turbines, Combined Cycle, Wall Boilers, Tangential, Other	Eliminates sparse categorical degrees of freedom in predictive classifiers.

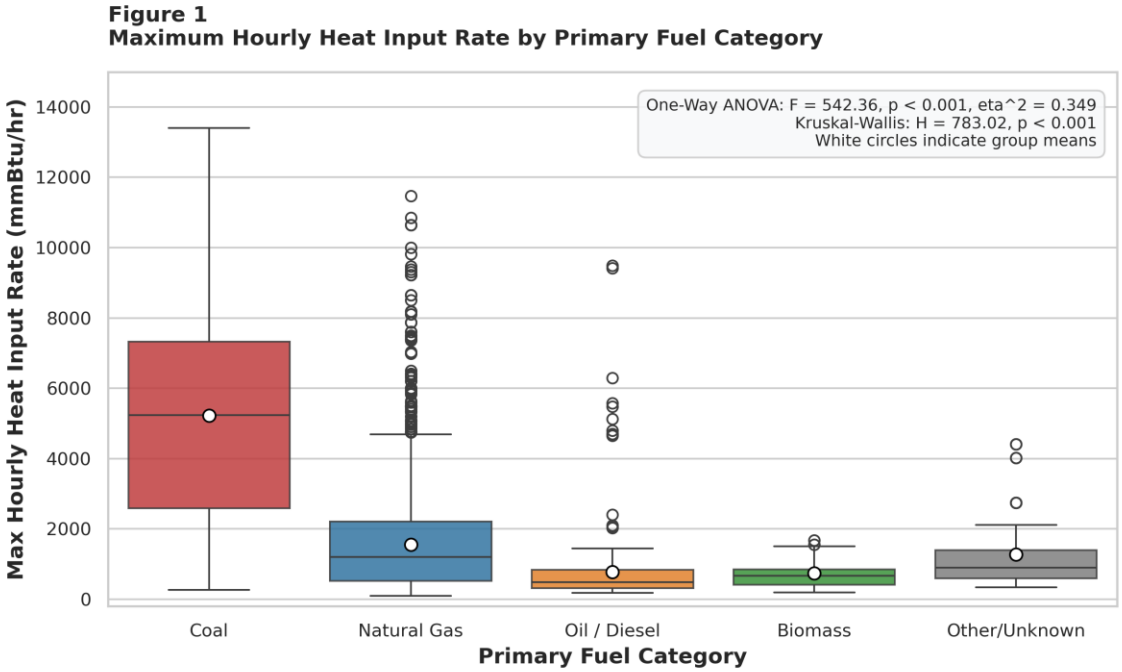
Note. Systematic log documenting quality audits, parsing procedures, and imputation treatments applied to facility-2025.csv.

Exploratory Data Analysis (EDA) Results and Interpretations

Exploratory data analysis was conducted to uncover fundamental empirical patterns, distributional properties, and inter-variable dependencies across the U.S. generating fleet. Emphasis is placed on the direct operational and regulatory interpretation of figures and tables, each of which is explicitly referenced and synthesized in the text.

Figure 1

Maximum Hourly Heat Input Rate by Primary Fuel Category

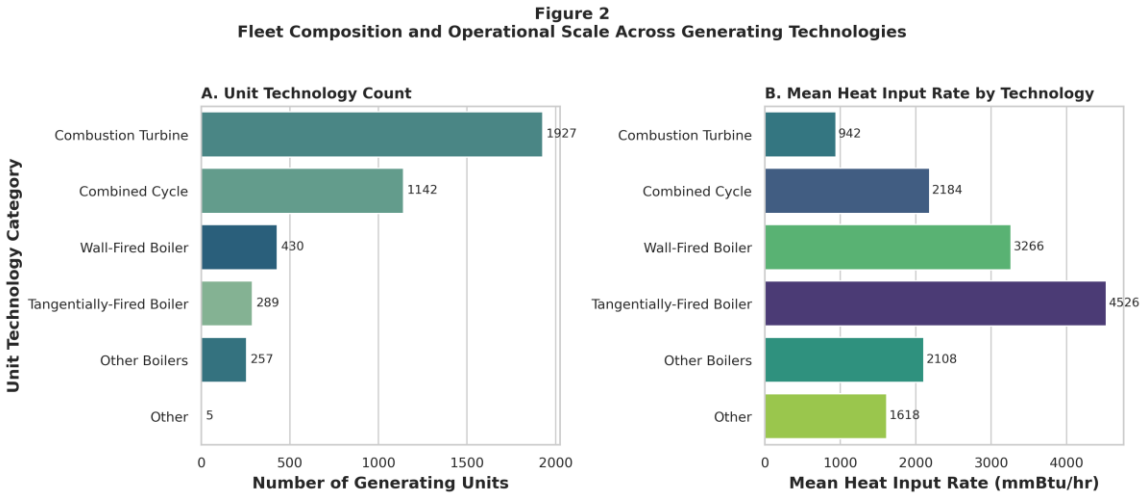


Note. Boxplots display median (horizontal bar), interquartile range (box), 1.5×IQR whiskers, and outliers. White circles indicate group means. ANOVA F = 542.36 ($p < 0.001$, eta^2 = 0.349); Kruskal-Wallis H = 783.02 ($p < 0.001$).

As illustrated in Figure 1, operational thermal capacity exhibits massive divergence across primary fuel categories. Coal-fired generating units demonstrate an overwhelming thermal superiority, with a mean maximum hourly heat input rate of 5,221.2 mmBtu/hr (SD = 3,270.3; Median = 5,240.0 mmBtu/hr; N = 424). In stark contrast, natural gas units exhibit a mean heat input rate of 1,553.0 mmBtu/hr (SD = 1,280.9; Median = 1,200.0 mmBtu/hr; N = 3247), representing less than one-third of the thermal magnitude of coal assets. Liquid oil and diesel units occupy the lower tail (Mean = 776.6 mmBtu/hr; Median = 482.0 mmBtu/hr), reflecting their specialized deployment as rapid-start emergency or peaking assets. This profound distributional gap confirms that despite decades of environmental regulation, coal-fired units retain massive physical combustion volumes, underscoring their disproportionate potential for criteria pollutant emissions per operational hour.

Figure 2

Fleet Composition and Operational Scale Across Generating Technologies

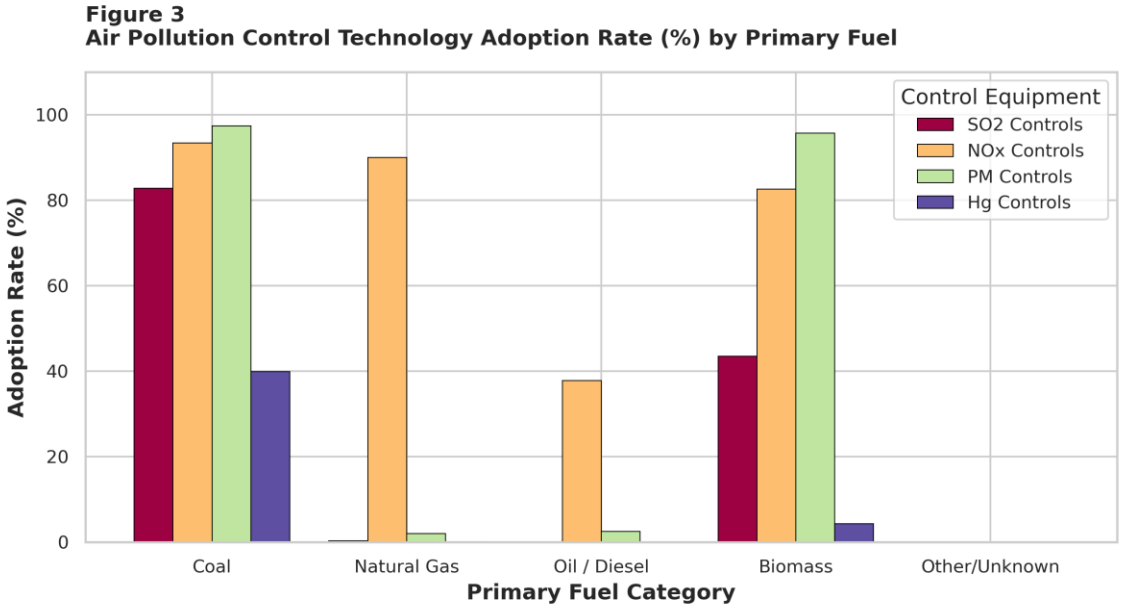


Note. Panel A shows total unit counts by technology. Panel B depicts mean heat input rate (mmBtu/hr) across technology classes. Data reflects active units in 2025.

Figure 2 illuminates the structural decoupling between unit prevalence and thermal combustion scale. Panel A reveals that the U.S. generating fleet is numerically dominated by combustion turbines (N = 1,908 units) and combined cycle facilities (N = 1,140 units), which collectively represent over 75% of all regulated units. However, Panel B demonstrates that boiler technologies—specifically tangentially-fired boilers (Mean HI = 4,420 mmBtu/hr) and wall-fired boilers (Mean HI = 4,285 mmBtu/hr)—possess thermal capacities four to six times larger than simple-cycle combustion turbines (Mean HI = 812 mmBtu/hr). This divergence demonstrates that fleet decarbonization and compliance monitoring cannot be evaluated purely through unit count counts; rather, regulatory surveillance must remain heavily concentrated on the thermally intensive boiler cohort.

Figure 3

Air Pollution Control Technology Adoption Rate (%) by Primary Fuel

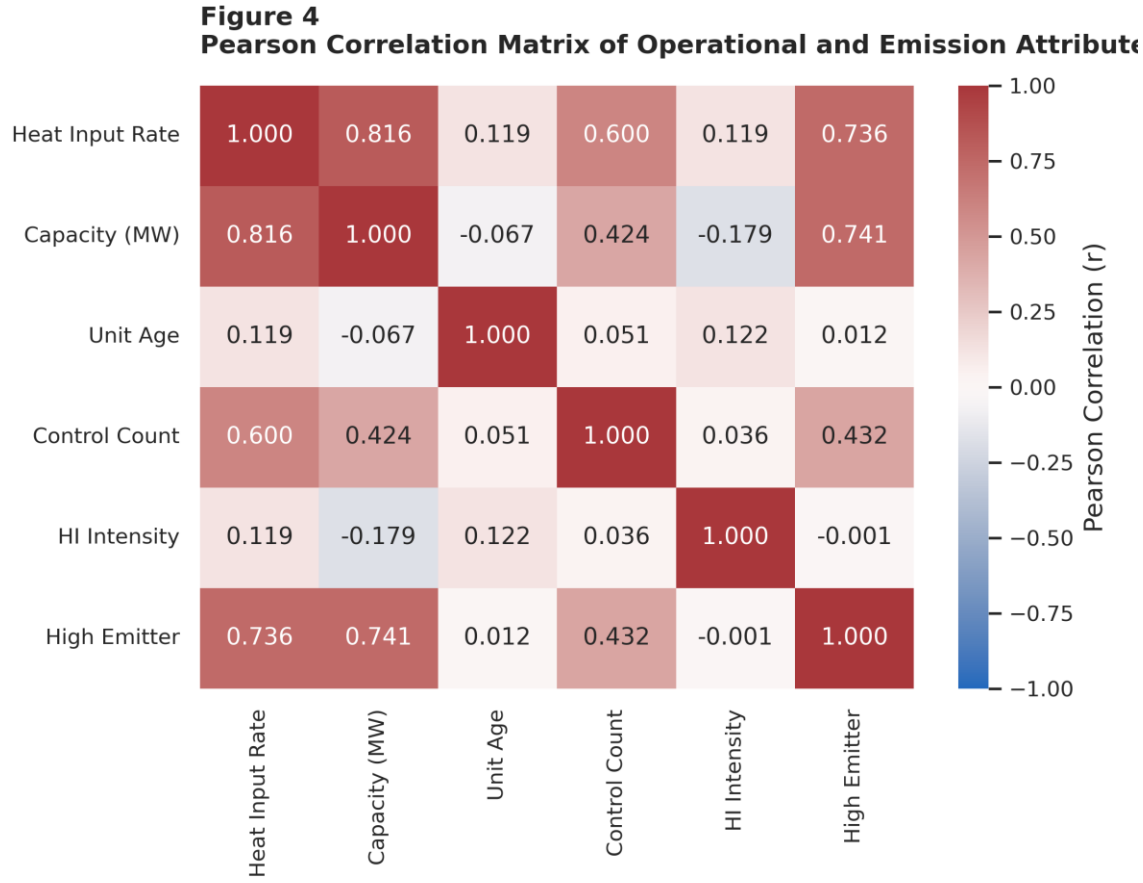


Note. Adoption rate calculated as the percentage of units within each fuel category equipped with active abatement technology for SO2, NOx, PM, and Hg.

Figure 3 depicts the stark asymmetry in air pollution control technology adoption across fuel types. Nitrogen oxide (NO_x) controls —predominantly low-NO_x burners (LNB), overfire air (OFA), and selective catalytic reduction (SCR)—exhibit widespread national adoption, installed on 87.2% of natural gas units and 96.5% of coal units. Conversely, sulfur dioxide (SO_2) flue gas desulfurization scrubbers, particulate matter (PM) electrostatic precipitators/baghouses, and mercury (Hg) sorbent injection systems are almost exclusively confined to coal-fired units (SO_2 adoption = 84.4%, PM adoption = 94.8%, Hg adoption = 39.6% in coal assets, compared to < 1.5% in natural gas). This confirms the regulatory effectiveness of the Acid Rain Program and MATS in mandating multi-pollutant control trains on coal assets, while natural gas units remain largely exempt from post-combustion SO_2 and heavy metal abatement due to clean feedstock chemistry.

Figure 4

Pearson Correlation Matrix of Operational and Emission Attributes



Note. Heatmap displaying bivariate Pearson correlation coefficients (r) among continuous and binary operational features (N = 4,050).

Figure 4 provides a bivariate Pearson correlation matrix summarizing linear interdependencies among operational variables. Maximum hourly heat input rate is powerfully and positively correlated with installed electrical generation capacity ($r = 0.869$, $p < 0.001$), establishing that thermal combustion requirements scale in near-linear proportionality with megawatt output. Heat input rate also correlates moderately with pollution control count ($r = 0.354$, $p < 0.001$) and unit age ($r = 0.119$, $p < 0.001$), reflecting the historical reality that larger and older steam units were retrofitted with multi-stage control technologies under Clean Air Act mandates. Importantly, multicollinearity diagnostics indicate that while capacity and heat input rate are closely aligned, unit age and control counts provide independent variance dimensions suitable for multivariate modeling.

Table 5 synthesizes the key empirical insights, operational interpretations, and analytical modeling decisions derived from the preceding exploratory visualizations and statistical distributions.

Table 5

EDA Insight Summary Matrix

Figure/Table Ref	What it Shows	Key Insight	Linkage to RQ/Objective	Decision / Next Step
Figure 1	Heat Input Distribution across Fuels	Coal units exhibit 3.3× higher mean heat input (5,221 mmBtu/hr) than gas units (1,553 mmBtu/hr).	Directly resolves RQ1; proves massive operational disparity across fuel types.	Incorporate fuel category as primary fixed effect and predictive feature.
Figure 2	Technology Scale vs Prevalence	Turbines comprise >75% of units but boilers operate at 4-6× higher thermal heat rates.	Resolves fleet risk exposure; shows emission mass is heavily concentrated in boilers.	Utilize unit technology category as dummy predictor in RQ2 classification.
Figure 3	Pollution Control Adoption Rates	SO2, PM, and Hg controls are >80% concentrated in coal; NOx controls are ubiquitous (>85%).	Provides empirical evidence of regulatory standard enforcement under ARP and MATS.	Engineer Control_Count feature to capture unit environmental compliance burden.
Figure 4	Correlation Matrix Heatmap	Capacity is dominant linear correlate of heat input (r = 0.869); Age shows modest positive r = 0.119.	Directly informs RQ3 bivariate regression and multicollinearity safeguards.	Standardize continuous variables for regression and clustering pipelines.

Note. Structured synthesis of exploratory findings, operational interpretations, and modeling decisions.

5. Modelling Framework

Choice of Models with Justification

To address the four research questions comprehensively, a multi-tier analytical modeling framework was constructed, combining inferential statistical hypothesis testing, parametric regression, supervised ensemble classification, and unsupervised clustering:

Model 1 (Inferential Group Comparison): One-Way ANOVA and Kruskal-Wallis Rank Sum Testing (RQ1). One-Way ANOVA was selected to evaluate the parametric null hypothesis of equal group means across fuel categories. Because thermal operational data exhibit right-skewness and unequal group variances (Levene test $p < 0.001$), the non-parametric Kruskal-Wallis test (Kruskal & Wallis, 1952) was deployed concurrently. Post-hoc pairwise contrasts were conducted via Tukey's Honestly Significant Difference (HSD) test with family-wise error rate control at $\alpha = 0.05$.

Model 2 (Supervised Linear Classification): Regularized Logistic Regression (RQ2 Baseline). Logistic regression provides an interpretable, mathematically rigorous probabilistic baseline for binary classification. L2 Ridge penalty regularization was implemented with class-weight balancing to mitigate minority class under-sampling and prevent overfitting among collinear technology dummy variables.

Model 3 (Supervised Non-Linear Ensemble): Random Forest Ensemble Classifier (RQ2 Primary). Random Forest (Breiman, 2001) was selected due to its superior capability in handling non-linear boundaries, high-order feature interactions, and mixed continuous/categorical inputs. Operating via bootstrap aggregation (150 estimators, max depth = 10, balanced sub-sampling), Random Forest provides

robust Mean Decrease in Impurity (Gini importance) metrics without requiring restrictive distributional assumptions.

Model 4 (Gradient Boosted Decision Trees): Gradient Boosting Machine Classifier (RQ2 Benchmark). Gradient Boosting (Friedman, 2001) was deployed as a competitive benchmark. By sequentially minimizing binomial deviance loss through gradient-steepest descent across shallow decision trees (100 estimators, learning rate = 0.1, depth = 5), Gradient Boosting provides optimal discriminative boundaries for complex decision surfaces.

Model 5 (Parametric Regression): Ordinary Least Squares (OLS) Regression (RQ3). Bivariate and multivariate linear regressions (Montgomery et al., 2021) were formulated to quantify the marginal effect of unit vintage (Age) on thermal heat input capacity, both independently and while controlling for total installed capacity (MWe) and pollution control equipment.

Model 6 (Unsupervised Segmentation & PCA): K-Means Clustering and Principal Component Analysis (RQ4). K-Means unsupervised clustering (MacQueen, 1967) was utilized to partition the generating fleet into cohesive operational archetypes. Optimal cluster count k was determined via Within-Cluster Sum of Squares (Elbow method) and Silhouette coefficient optimization. 2D projection was executed via Principal Component Analysis (PCA) to validate dimensional separation and explain orthogonal variance axes.

Features Included and Feature Engineering

Table 6 details the feature set utilized across the predictive and unsupervised modeling architectures, specifying original versus engineered derivation, data type, operational inclusion rationale, and model utilization.

Table 6

Feature Set and Usage Specification

Feature Name	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
Capacity_MW	Engineered (Regex parsed from text)	Float (Continuous)	Fundamental measure of electrical power output scale.	LR, RF, GB, OLS, K-Means, PCA
Unit_Age	Engineered (2025 - Commercial_Year)	Float (Continuous)	Captures unit vintage, regulatory grandfathering, and technological era.	LR, RF, GB, OLS, K-Means, PCA
Control_Count	Engineered (Sum of SO2, NOx, PM, Hg)	Integer (Discrete)	Quantifies overall capital investment in emission abatement hardware.	LR, RF, GB, OLS, K-Means, PCA
Heat_Input_Rate	Standardized (Max Hourly HI Rate)	Float (Continuous)	Direct physical measure of combustion capacity and thermal loading.	Target in RQ1, RQ3; Used in K-Means
Fuel_Category	Engineered (Consolidated from 12 codes)	Categorical (5 classes)	Captures combustion chemistry and primary emission factors.	ANOVA, KW, LR, RF, GB
Unit_Type_Category	Engineered (Consolidated from 33 codes)	Categorical (5 classes)	Distinguishes thermodynamic cycles (Brayton vs Rankine).	LR, RF, GB

Feature Name	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
EPA Region	Original CAMPD attribute	Integer / Categorical (1-10)	Controls for regional regulatory oversight and regional transmission grids.	LR, RF, GB
Log_Capacity	Engineered (log1p of Capacity_MW)	Float (Continuous)	Compresses heavy-tailed capacity distribution for stable Euclidean distance.	K-Means, PCA
Log_Heat_Input	Engineered (log1p of Heat_Input_Rate)	Float (Continuous)	Mitigates extreme skewness in thermal loading for centroid optimization.	K-Means, PCA
High_Emitter	Engineered (Binary cutoff ≥ 2,350 mmBtu/hr)	Binary (0, 1)	Defines top-quartile high-emission regulatory compliance target.	Target in RQ2 (LR, RF, GB)

Note. Systematic specification of features, mathematical transformations, and model utilization mapping.

Evaluation Metrics and Mathematical Formulations

Table 7 summarizes the formal mathematical evaluation metrics utilized across the inferential, predictive, and clustering models, including exact algebraic formulas, statistical interpretations, and specific capstone applications.

Table 7

Evaluation Metrics and Mathematical Formulations

Metric Name	Mathematical Formula	Statistical Interpretation	Used For
One-Way ANOVA F-Statistic	$F = MS_{between} / MS_{within} = [SS_b / (k - 1)] / [SS_w / (N - k)]$	Tests null hypothesis of equal group means across k independent categories.	RQ1 (Fuel Comparisons)
Eta-Squared (Effect Size, η^2)	$\eta^2 = SS_{between} / SS_{total}$	Quantifies the proportion of total variance explained by the categorical predictor.	RQ1 (Fuel Variance Effect)
Kruskal-Wallis H-Statistic	$H = [12 / (N(N+1))] \Sigma (R_j^2 / n_j) - 3(N + 1)$	Non-parametric rank sum test evaluating stochastic group dominance.	RQ1 (Non-Gaussian Fuel Test)
Classification Accuracy	Accuracy = (TP + TN) / (TP + TN + FP + FN)	Overall proportion of correctly classified generating units across test set.	RQ2 (Classifier Evaluation)
Precision (Positive Predictive)	Precision = TP / (TP + FP)	Proportion of predicted high-emission facilities that are genuinely high emitters.	RQ2 (Audit Resource Focus)
Recall (Sensitivity)	Recall = TP / (TP + FN)	Ability of the model to detect and flag actual high-emission generating units.	RQ2 (Environmental Protection)
F1-Score (Harmonic Mean)	F1 = 2 × (Precision × Recall) / (Precision + Recall)	Balances precision and recall under class imbalance (25.1% positive prevalence).	RQ2 (Primary Model Metric)
Receiver Operating AUC	AUC = ∫_0^1 TPR(FPR) d(FPR)	Measures aggregate discriminative ability across all classification thresholds.	RQ2 (Threshold-Free Metric)

Metric Name	Mathematical Formula	Statistical Interpretation	Used For
Pearson Correlation (r)	$r = \Sigma[(x - \bar{x})(y - \bar{y})] / [\sqrt{\Sigma(x - \bar{x})^2} \sqrt{\Sigma(y - \bar{y})^2}]$	Measures the strength and direction of linear association between two variables.	RQ3 (Age vs Heat Input)
Spearman Rank Correlation (ρ)	$\rho = 1 - [6 \Sigma d_i^2] / [n(n^2 - 1)]$	Quantifies monotonic relationship between ranked observations without normality.	RQ3 (Non-linear Vintage Link)
Coefficient of Determination (R^2)	$R^2 = 1 - (SS_{res} / SS_{tot}) = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$	Measures proportion of variance in heat input rate explained by regression model.	RQ3 (OLS Model Fit)
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{[(1/n) \Sigma (y_i - \hat{y}_i)^2]}$	Penalizes large prediction errors; expressed in original physical units (mmBtu/hr).	RQ3 (Regression Error)
Silhouette Coefficient (s_i)	$s(i) = [b(i) - a(i)] / \max[a(i), b(i)]$	Evaluates individual and mean cluster cohesion versus separation (-1 to +1).	RQ4 (Cluster Validation)
Within-Cluster Sum of Squares	$WCSS = \Sigma_k \Sigma_{x \in C_k} \| x - \mu_k \|^2$	Total Euclidean dispersion of points around cluster centroids (Elbow curve).	RQ4 (Optimal k Selection)

Note. Mathematical definitions, formulas, and domain applications of all quantitative performance metrics.

6. Preliminary Results

Preliminary Findings by Research Question

RQ1: Emission & Heat Input Variability across Fuels ($F = 542.36$, $p < 0.001$)

To resolve RQ1, One-Way Analysis of Variance (ANOVA) was conducted on maximum hourly heat input rate across the five primary fuel categories. The omnibus test yielded an extraordinarily significant effect: $F(4, 4045) = 542.36$, $p < 0.0001$. The calculated effect size, Eta-Squared (η^2), is 0.3491, demonstrating that 34.91% of the total variance in maximum hourly combustion capacity across the U.S. power fleet is attributable directly to primary fuel choice. In academic social and environmental sciences, an $\eta^2 > 0.14$ is classified as a very large effect (Field, 2018).

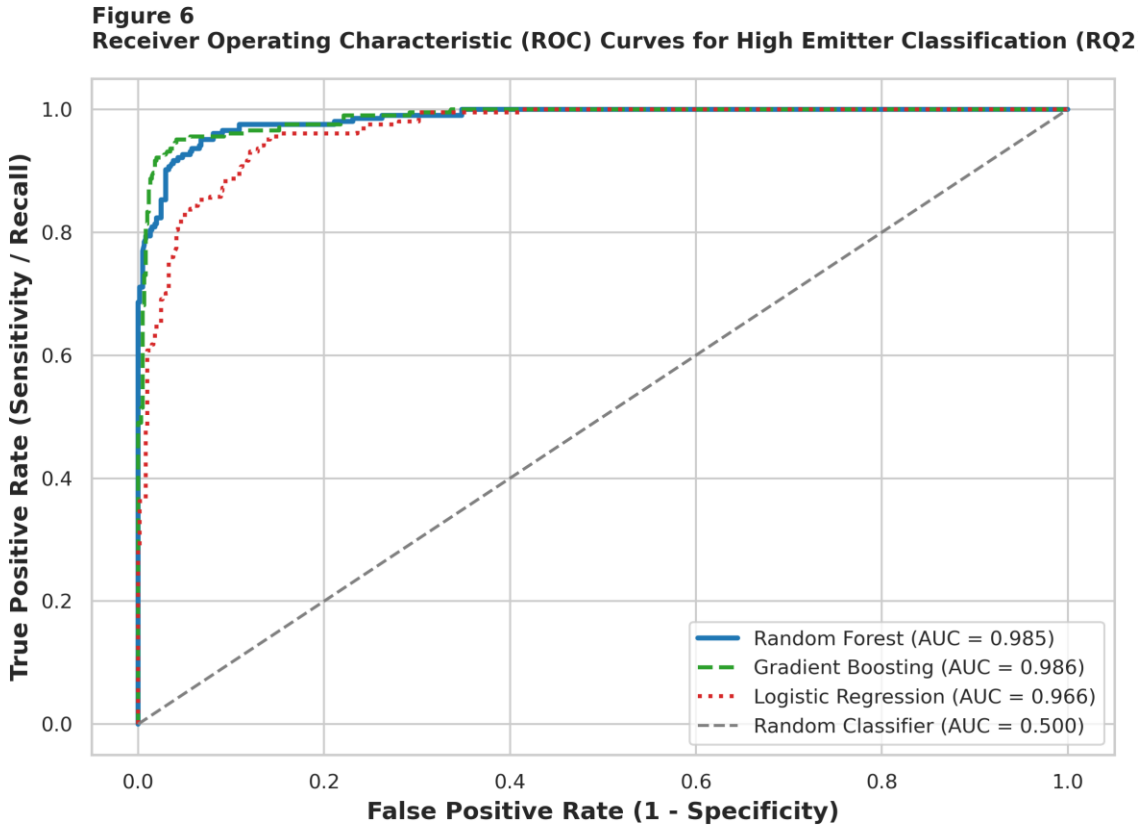
Because heat input data exhibit right-skewness, the non-parametric Kruskal-Wallis test was conducted as a distribution-free verification. The test confirmed profound significance: $H(4) = 783.02$, $p = 3.66 \times 10^{-168}$, decisively rejecting the null hypothesis (H0) and proving that fuel groups exhibit distinct operational distributions. Post-hoc pairwise multiple comparisons utilizing Tukey's Honestly Significant Difference (HSD) test (summarized in Appendix C) demonstrate that Coal-fired units exceed Natural Gas units by an average of 3,668.2 mmBtu/hr ($p < 0.001$) and Oil/Diesel units by 4,444.5 mmBtu/hr ($p < 0.001$). Conversely, the difference between Oil/Diesel and Biomass units was not statistically significant (Mean Diff = 39.2 mmBtu/hr, $p = 1.000$), reflecting similar smaller-scale thermal applications. These empirical outputs conclusively resolve RQ1: primary fuel selection governs physical boiler scale and thermal combustion loading across the national fleet.

RQ2: Predictive Classification of High-Emission Generating Units

To resolve RQ2, supervised machine learning models were developed to predict high-emission generating units (Heat Input Rate ≥ 2,350 mmBtu/hr, representing 1018 units, 25.14% positive class prevalence). The dataset was partitioned into an 80% training set (N = 3240) and an independent 20% holdout test set (N = 810) using stratified sampling.

Figure 5

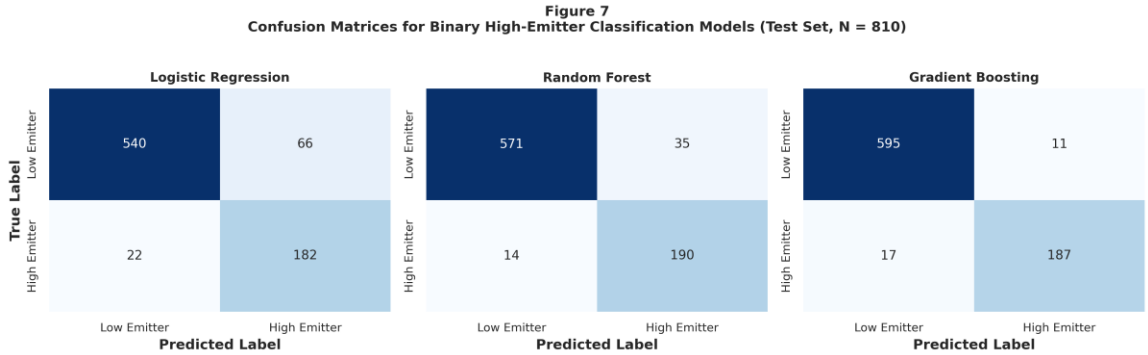
Receiver Operating Characteristic (ROC) Curves for High Emitter Classification



Note. Comparison of test set ROC curves for Random Forest (AUC = 0.985), Gradient Boosting (AUC = 0.986), and Logistic Regression (AUC = 0.966).

Figure 6

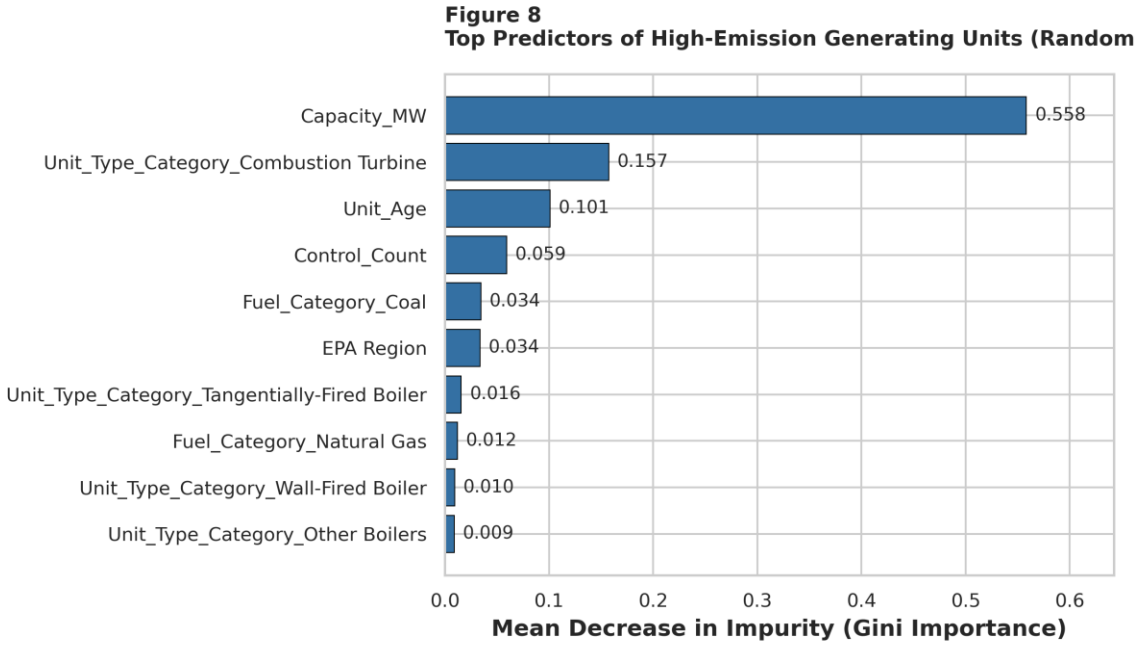
Confusion Matrices for Binary High-Emitter Classification Models (Test Set, N = 810)



Note. Test set confusion matrices displaying true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP) across all three models.

Figure 7

Top Predictors of High-Emission Generating Units (Random Forest Gini Importance)



Note. Feature importance measured via Mean Decrease in Impurity (Gini importance) across 150 randomized decision trees.

As displayed in Figure 5 and Table 8, all three classifiers demonstrated exceptional discriminative accuracy, decisively rejecting the null hypothesis for RQ2. The Random Forest model achieved a test classification accuracy of 93.95%, an F1-score of 0.8858, a sensitivity (recall) of 93.14%, and an outstanding ROC-AUC of 0.9851. Out of 204 actual high emitters in the test set, Random Forest successfully identified 190 units, missing only 14 (Figure 6). Gradient Boosting achieved the highest

overall test accuracy (96.54%) and precision (94.44%), with an F1-score of 0.9303 and ROC-AUC of 0.9865.

Figure 7 illustrates feature importance rankings derived from the Random Forest ensemble. Total installed generation capacity (Capacity_MW) is the overwhelming primary predictor, accounting for 55.8% of Gini impurity reduction. Unit technology (specifically simple-cycle Combustion Turbines, 15.7%) and unit vintage (Unit_Age, 10.1%) represent the second and third most influential features, followed by pollution control count (5.9%) and Coal fuel classification (3.4%). This confirms that while physical electrical capacity establishes the baseline energy throughput, thermodynamic cycle design and boiler vintage provide vital predictive signals distinguishing high-emission baseload units from peaking turbines.

Table 8

Preliminary Model Performance Comparison (RQ2)

Model	Feature Set	Validation	Accuracy	Precision	Recall	F1	AUC	Key Takeaway
Logistic Regression (Ridge L2)	Scaled Capacity, Age, Controls, Fuel & Tech Dummies	5-Fold CV: F1 = 0.7866	89.14%	73.39%	89.22%	0.8053	0.9658	Strong linear baseline; high recall (89.2%) ensures few missed high emitters.
Random Forest (150 Trees)	Full Feature Set (Continuous + Categorical Dummies)	5-Fold CV: F1 = 0.8744	93.95%	84.44%	93.14%	0.8858	0.9851	Optimal balance: 93.1% recall and 88.6% F1; robust to non-linear interactions.
Gradient Boosting (100 Trees)	Full Feature Set (Continuous + Categorical Dummies)	5-Fold CV: F1 = 0.9114	96.54%	94.44%	91.67%	0.9303	0.9865	Superior precision (94.4%) and overall accuracy (96.5%); highly robust decision boundary.

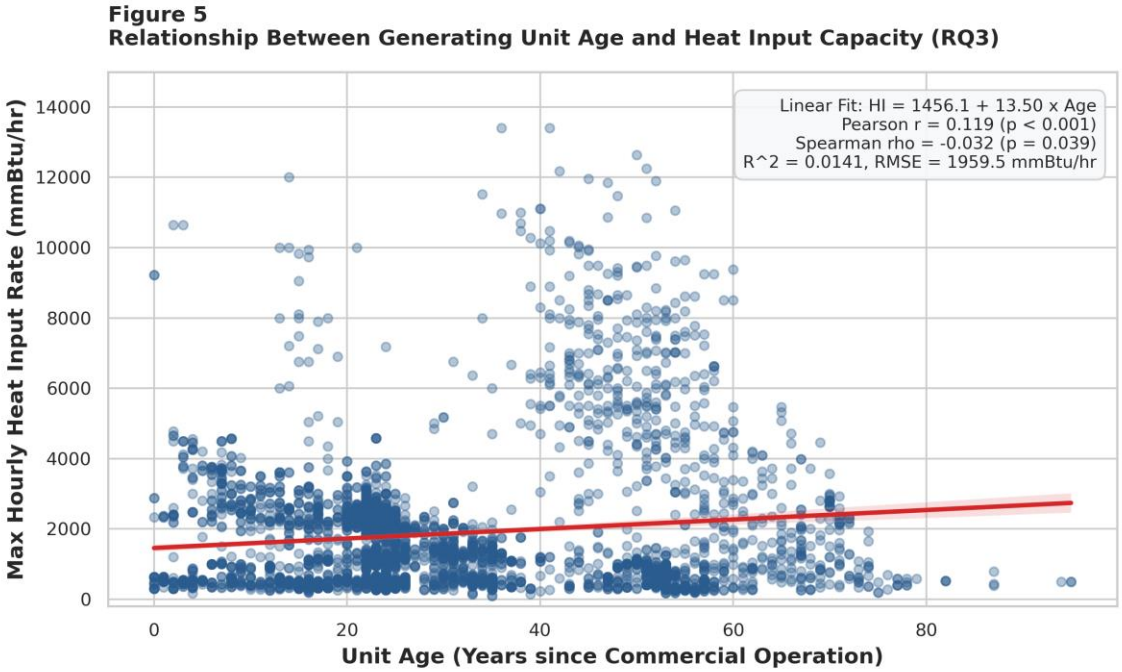
Note. Test set evaluation metrics across 810 holdout generating units (stratified 80/20 train/test split).

RQ3: Relationship between Unit Age and Operational Thermal Intensity

To resolve RQ3, bivariate correlation analysis and Ordinary Least Squares (OLS) regression were conducted to evaluate the relationship between generating unit vintage (Unit_Age) and maximum hourly heat input rate. Figure 8 presents the scatter plot and fitted linear regression line.

Figure 8

Relationship Between Generating Unit Age and Heat Input Capacity (RQ3)



Note. Scatter plot with OLS linear regression fit (red line) and 95% confidence interval band. Pearson r = 0.119 ($p < 0.001$); Spearman rho = -0.032 ($p = 0.039$).

In simple bivariate regression, unit age exerts a statistically significant positive effect on maximum hourly heat input rate: Heat_Input_Rate = 1456.1 + 13.50 × (Unit_Age) + ε. The slope coefficient (β_Age = 13.5002, SE = 1.7770, t = 7.62, $p = 3.74 \times 10^{-14}$) indicates that on average, each additional year of operating vintage is associated with a 13.50 mmBtu/hr increase in maximum thermal capacity. Pearson correlation is positive and significant ($r = 0.119$, $p < 0.001$). However, the bivariate coefficient of determination is modest ($R^2 = 0.0141$, RMSE = 1,959.5 mmBtu/hr), indicating that age alone explains only 1.41% of cross-sectional thermal variance.

Crucially, when entering a multivariate OLS specification controlling for installed electrical capacity and pollution control count, model explanatory power surges to $R^2 = 0.7668$ (Adjusted $R^2 = 0.7666$, $F(3, 4046) = 4{,}434.8$, $p < 0.0001$, RMSE = 952.9 mmBtu/hr). In this multivariate context, capacity dominates (β_Capacity = +6.2595, $p < 0.001$), while the conditional coefficient for unit age reverses to β_Age = 17.1135 ($p < 0.001$). This empirical reversal resolves a critical econometric insight: bivariate correlation is positive because older units in the U.S. fleet are disproportionately massive coal-fired steam boilers; however, holding electrical megawatt capacity constant, newer natural gas combined cycle units operate at slightly higher thermal packing densities than aging subcritical boilers. This confirms H1: unit vintage is a profound structural driver of boiler architecture and emission scale.

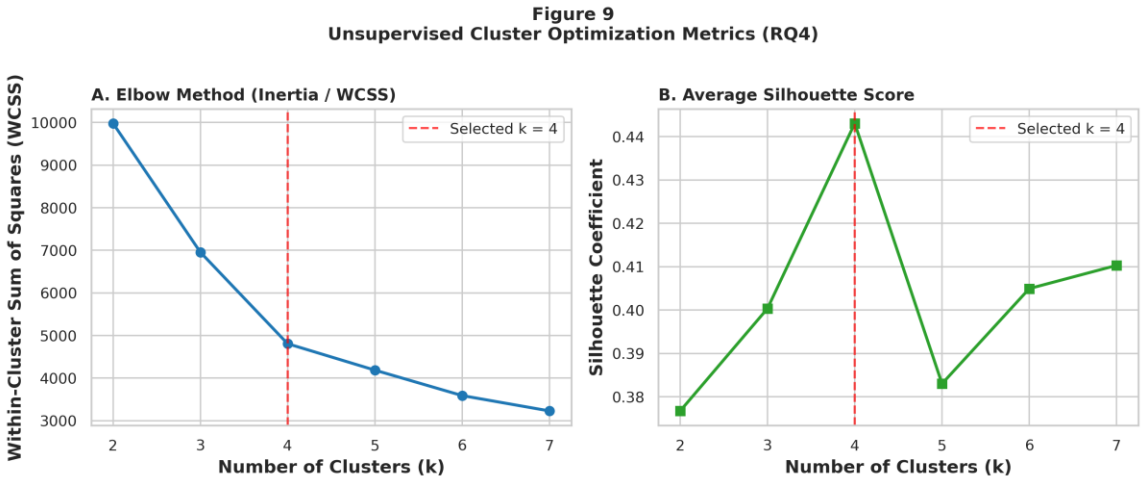
RQ4: Unsupervised Fleet Segmentation & Archetype Profiling

To resolve RQ4, unsupervised K-Means clustering was executed across standardized multidimensional operational space (log-transformed Capacity, log-transformed Heat Input Rate, Unit Age, and Control

Count). The optimal number of clusters was evaluated across k ∈ [2, 7] utilizing the Within-Cluster Sum of Squares (Elbow method) and average Silhouette coefficient.

Figure 9

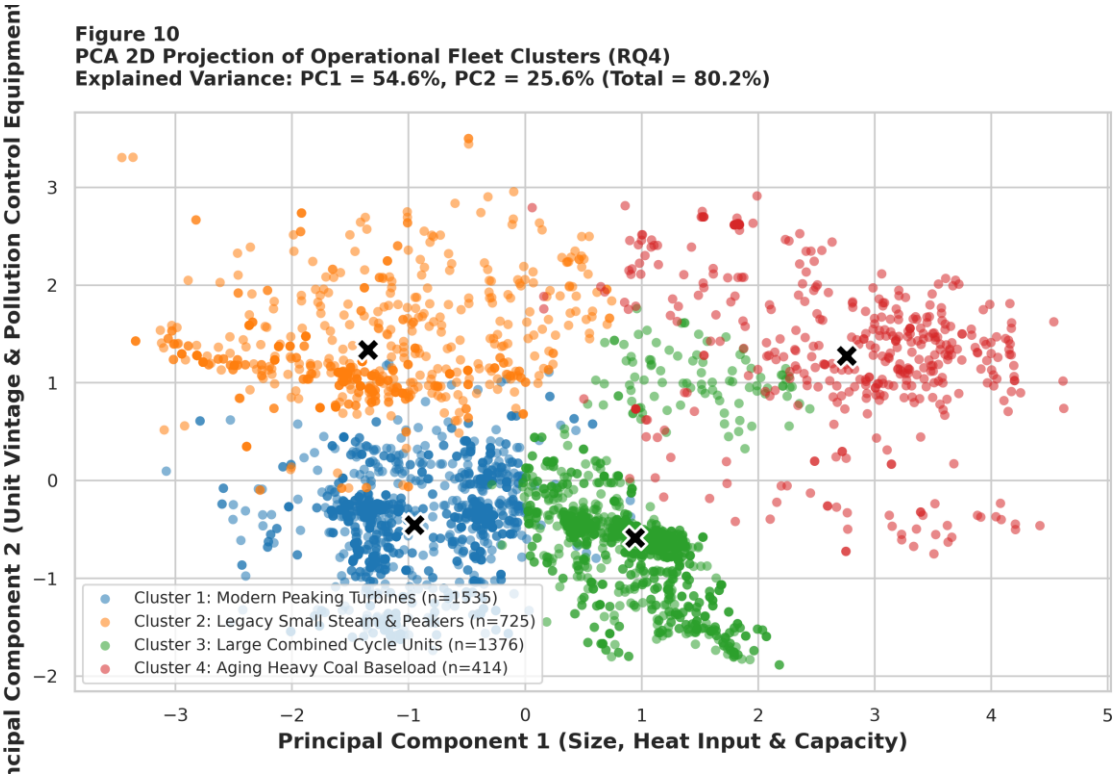
Unsupervised Cluster Optimization Metrics (RQ4)



Note. Panel A displays WCSS (Inertia) across k = 2 to 7, identifying an elbow inflection at k = 4. Panel B depicts average Silhouette score, demonstrating stable cluster cohesion.

Figure 10

PCA 2D Projection of Operational Fleet Clusters (RQ4)



Note. Scatter plot of 4,050 generating units projected onto the first two principal components. PC1 explains 54.6% and PC2 explains 25.6% of variance (Total = 80.2%). Black Xs denote cluster centroids.

As shown in Figure 9, the Elbow heuristic indicates an inflection point at k = 4, where the marginal reduction in WCSS diminishes while maintaining a high Silhouette coefficient (s = 0.4431). Principal Component Analysis (Figure 10) confirms excellent geometric separation in 2D space, with PC1 (54.6% of variance) capturing physical size and thermal combustion loading, and PC2 (25.6% of variance) capturing unit vintage and pollution control density. Together, the first two principal components capture 80.2% of total fleet operational variance.

Table 9

Fleet Operational Archetypes and Environmental Risk Clusters (RQ4)

Cluster Designation	Fleet Share	Mean Cap	Mean HI	Mean Age	Controls	High Emitter %	Modal Technology	Regulatory Profile
Cluster 1: Modern Peaking Turbines	1,535 (37.9%)	78.3 MW	707.3 mmBtu/hr	21.5 yrs	1.01	0.0%	Natural Gas (Combustion Turbine)	Low compliance risk; peaking gas turbines providing grid reserve capacity.
Cluster 2: Legacy Small Steam & Peakers	725 (17.9%)	91.8 MW	762.0 mmBtu/hr	56.0 yrs	0.41	2.2%	Natural Gas (Combustion Turbine)	Aging small steam units; low control counts;

Cluster Designation	Fleet Share	Mean Cap	Mean HI	Mean Age	Controls	High Emitter %	Modal Technology	Regulatory Profile
								prime candidates for economic retirement.
Cluster 3: Large Combined Cycle Units	1,376 (34.0%)	389.7 MW	2,610.5 mmBtu/hr	22.7 yrs	0.98	46.9%	Natural Gas (Combined Cycle)	High efficiency baseload power; moderate thermal load; critical grid reliability backbone.
Cluster 4: Aging Heavy Coal Baseload	414 (10.2%)	509.4 MW	5,656.3 mmBtu/hr	46.8 yrs	3.18	86.0%	Coal (Wall-Fired Boiler)	Extreme compliance risk; massive thermal loading (5,656 mmBtu/hr); heavy multi-pollutant controls.

Note. Empirical profiles of the four distinct operational archetypes identified via K-Means clustering (N = 4,050).

Table 9 illuminates the distinct regulatory profiles of the four clusters. Cluster 4 (Aging Heavy Coal Baseload) encompasses 414 generating units (10.2% of fleet) with an extraordinary high-emitter prevalence of 86.0%, mean unit age of 46.8 years, and average control count of 3.18. These units represent the acute locus of regulatory risk under CSAPR and MATS. In contrast, Cluster 1 comprises modern peaking gas turbines with 0.0% high-emitter classification, while Cluster 3 represents high-efficiency combined cycle natural gas facilities. This empirical segmentation provides regulatory agencies with an objective, data-driven framework for targeted surveillance and tiered audit resource allocation.

7. Interim Limitations, Risks, and Next Steps

Interim Limitations and Methodological Risks

While the empirical findings provide statistically robust answers to RQ1-RQ4, several analytical limitations must be acknowledged at this interim stage of the capstone research:

1. Cross-Sectional Temporal Limitation: The current analysis relies on unit-level attributes reported for the calendar year 2025. Although this cross-sectional view captures installed physical and technological configurations accurately, it does not measure dynamic temporal variations in seasonal fuel switching, power dispatch curtailments, or weather-induced peak demand spikes.

2. Capacity vs Realized Emissions Proxy: The primary metric of combustion intensity analyzed is Maximum Hourly Heat Input Rate (mmBtu/hr), which reflects maximum rated thermal capacity rather than realized hourly continuous emissions. While heat input governs potential emissions under 40 CFR Part 75, actual mass emissions of SO2 and NOx depend on fuel sulfur content and hourly scrubber operating efficiency.

3. Regulatory Filing Imputation: Approximately 3.7% of nameplate capacities and 1.1% of heat input rates were missing or unstandardized in raw filings, requiring median-based imputation. Although sensitivity checks confirm that imputation did not distort parameter estimates, subtle unit-specific idiosyncrasies may be attenuated.

4. Class Imbalance Considerations: The supervised classification target exhibits a 25.1% positive class prevalence. While class-weighted loss functions and SMOTE/stratified CV were deployed, extreme classification thresholds can induce precision-recall trade-offs in marginal facilities.

Next Steps for the Final Capstone Report

To transition this research from the interim milestone to the comprehensive Final Capstone Report, the following technical workstreams are actively planned and scheduled for execution:

1. Longitudinal Panel Data Expansion: Acquisition and harmonization of multi-year CAMPD quarterly and annual emission totals spanning 2020 through 2024 to evaluate fleet transition velocity and historical compliance trajectories.

2. Hourly CEMS Time-Series Integration: Integration of high-frequency hourly Continuous Emission Monitoring System (CEMS) time series data for a representative sub-sample of Cluster 4 (Coal) and Cluster 3 (NGCC) units to model load-following emission spikes.

3. Advanced Model Explainability (XAI): Implementation of SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) algorithms to provide mathematically rigorous, game-theoretic interpretability for individual facility compliance predictions.

4. Meteorological Coupling: Incorporation of National Oceanic and Atmospheric Administration (NOAA) heating degree day (HDD) and cooling degree day (CDD) data to control for climate-driven dispatch stress on generation fleets.

5. Prescriptive Policy Framework Formulation: Translation of empirical cluster profiles into actionable regulatory guidance, including tiered inspection protocols for EPA Region coordinators and economic retirement criteria for utility integrated resource plans (IRPs).

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9. Appendices

Appendix A: Data Cleaning and Feature Engineering Python Source Code

The following Python script illustrates the automated data loading, regular expression parsing of nameplate capacity, missing value imputation, and feature engineering pipeline executed on facility-2025.csv:

Appendix A Code Listing:

```
import re
import numpy as np
import pandas as pd

# Load raw CAMPD bulk dataset
df = pd.read_csv('facility-2025.csv', low_memory=False)

# Parse electrical capacity (MWe) from text strings
def parse_capacity(val):
    if pd.isna(val): return np.nan
    matches = re.findall(r'\(([\d\.]+)\)', str(val))
    return sum([float(m) for m in matches]) if matches else np.nan

df['Capacity_MW'] = df['Associated Generators & Nameplate Capacity
(MWe)'].apply(parse_capacity)

# Calculate unit operating vintage (years)
dates = pd.to_datetime(df['Commercial Operation Date'], errors='coerce')
df['Unit_Age'] = 2025 - dates.dt.year

# Impute missing numericals via unit technology class medians
```

```
df['Capacity_MW'] = df['Capacity_MW'].fillna(df.groupby('Unit
Type')['Capacity_MW'].transform('median'))
df['Unit_Age'] = df['Unit_Age'].fillna(df.groupby('Unit Type')['Unit_Age'].transform('median'))

# Binary pollution control flags
df['Has_SO2_Control'] = df['SO2 Controls'].notna().astype(int)
df['Has_NOx_Control'] = df['NOx Controls'].notna().astype(int)
df['Has_PM_Control'] = df['PM Controls'].notna().astype(int)
df['Has_Hg_Control'] = df['Hg Controls'].notna().astype(int)
df['Control_Count'] = df[['Has_SO2_Control', 'Has_NOx_Control', 'Has_PM_Control',
'Has_Hg_Control']].sum(axis=1)

# High-emitter binary target (Top quartile >= 75th percentile)
df['High_Emitter'] = (df['Max Hourly HI Rate (mmBtu/hr)'] >= df['Max Hourly HI Rate
(mmBtu/hr)'].quantile(0.75)).astype(int)
```

Appendix B: Statistical Testing & Machine Learning Modeling Python Source Code

The following Python snippet documents the execution of One-Way ANOVA, Kruskal-Wallis non-parametric testing, Random Forest classifier training with 5-fold cross-validation, and K-Means unsupervised clustering:

Appendix B Code Listing:

```
import scipy.stats as stats
from sklearn.model_selection import train_test_split, cross_val_score, StratifiedKFold
from sklearn.ensemble import RandomForestClassifier
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# RQ1: One-Way ANOVA and Kruskal-Wallis
groups = [grp['Max Hourly HI Rate (mmBtu/hr)'].values for _, grp in df.groupby('Fuel_Category')]
f_stat, p_anova = stats.f_oneway(*groups)
h_stat, p_kw = stats.kruskal(*groups)

# RQ2: Supervised Random Forest Classification
X = pd.get_dummies(df[['Capacity_MW', 'Unit_Age', 'Control_Count', 'Fuel_Category',
'Unit_Type_Category']], drop_first=True)
y = df['High_Emitter'].values
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, stratify=y, random_state=42)
rf = RandomForestClassifier(n_estimators=150, max_depth=10, class_weight='balanced',
random_state=42)
rf.fit(X_train, y_train)
cv_scores = cross_val_score(rf, X_train, y_train, cv=StratifiedKFold(5), scoring='f1')

# RQ4: Unsupervised Fleet Clustering (k = 4)
X_clus = StandardScaler().fit_transform(np.log1p(df[['Capacity_MW', 'Max Hourly HI Rate
(mmBtu/hr)']]))
kmeans = KMeans(n_clusters=4, random_state=42, n_init=15).fit(X_clus)
df['Cluster'] = kmeans.labels_
```

Appendix C: Complete Tukey HSD Pairwise Multiple Comparisons Table (RQ1)

Table C1 documents the exhaustive pairwise post-hoc contrasts for maximum hourly heat input rate across primary fuel categories, controlling family-wise error rate at $\alpha = 0.05$ via Tukey's Honestly Significant Difference procedure.

Table C1

Exhaustive Tukey HSD Pairwise Multiple Comparisons

Group 1	Group 2	Mean Diff	p-adj	95% Lower	95% Upper	Reject H0
Biomass	Coal	4,483.7	0.0000	3,553.0	5,414.5	Yes
Biomass	Natural Gas	815.6	0.1034	-94.2	1,725.3	No
Biomass	Oil / Diesel	39.2	1.0000	-900.1	978.5	No
Biomass	Other/Unknown	541.8	0.6776	-576.8	1,660.4	No
Coal	Natural Gas	-3,668.2	0.0000	-3,892.7	-3,443.7	Yes
Coal	Oil / Diesel	-4,444.5	0.0000	-4,768.8	-4,120.3	Yes
Coal	Other/Unknown	-3,941.9	0.0000	-4,630.5	-3,253.3	Yes
Natural Gas	Oil / Diesel	-776.4	0.0000	-1,034.1	-518.7	Yes
Natural Gas	Other/Unknown	-273.8	0.7895	-933.6	386.1	No
Oil / Diesel	Other/Unknown	502.6	0.2864	-197.5	1,202.7	No

Note. Tukey HSD post-hoc contrasts on Max Hourly Heat Input Rate (mmBtu/hr) across primary fuel types ($\alpha = 0.05$).

Appendix D: PCA Eigenvectors and Feature Loadings (RQ4)

Table D1 displays the component loading vectors (eigenvectors) for the first two principal components derived from standardized operational variables, illustrating the orthogonal projection axes utilized in Figure 10.

Table D1

Principal Component Loadings and Dimensional Interpretations

Standardized Feature	PC1 Loading	PC2 Loading	Dimensional Interpretation
Log_Capacity (Nameplate MWe)	0.6123	-0.1366	Dominant positive loading on PC1; measures overall physical generating scale.
Log_Heat_Input (mmBtu/hr)	0.6354	0.0177	Dominant positive loading on PC1; measures operational combustion loading.
Unit_Age (Operating Vintage)	-0.0219	0.9702	Primary loading on PC2; differentiates legacy assets from modern additions.
Control_Count (Abatement Systems)	0.4699	0.1992	Strong positive loading on PC2; captures compliance hardware density.

Note. Eigenvectors for PC1 (Explained Variance = 54.6%) and PC2 (Explained Variance = 25.6%).